VideoSearcher Serverless Pipeline using OpenFaaS

Load Testing & Performance Evaluation

Soumyadeep Sharma

April 20, 2026

Introduction to OpenFaaS

What is OpenFaaS

OpenFaaS (Function as a Service) is an open-source framework for deploying serverless functions and microservices to **Kubernetes or Docker Swarm**. It enables developers to package any code, binary, or container image into a portable, auto-scaling function that responds to events.

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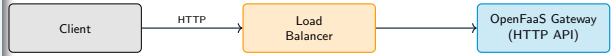

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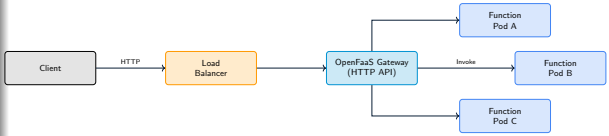

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- **Built-in Monitoring:** Integrates with Prometheus for metrics, logging, and performance tracking.

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- So instead of changing the code, I designed the entire infrastructure around it to make it cloud-compatible.

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- One of the biggest challenges I faced was that the code depends on a package called `aisprint`, which is not publicly available.
- So I had to find a way to make everything run smoothly without modifying or breaking the original implementation.

Pipeline Stages

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- Each of these steps was deployed as an independent OpenFaaS function.

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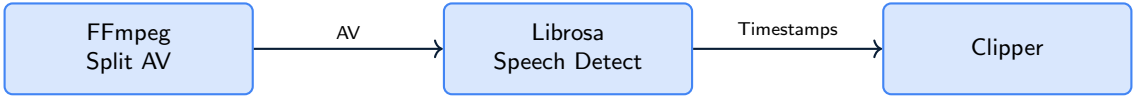

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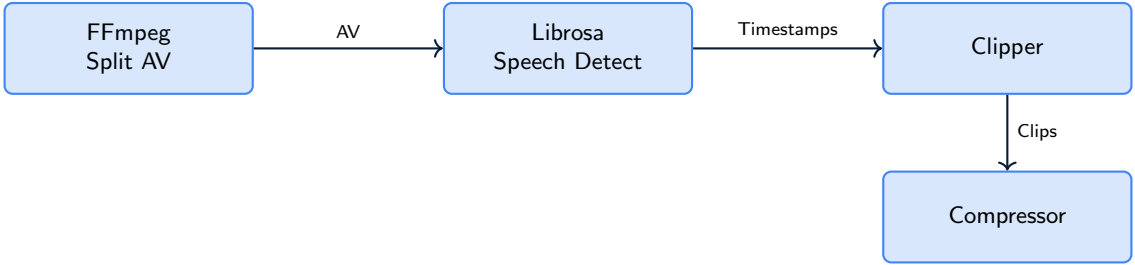

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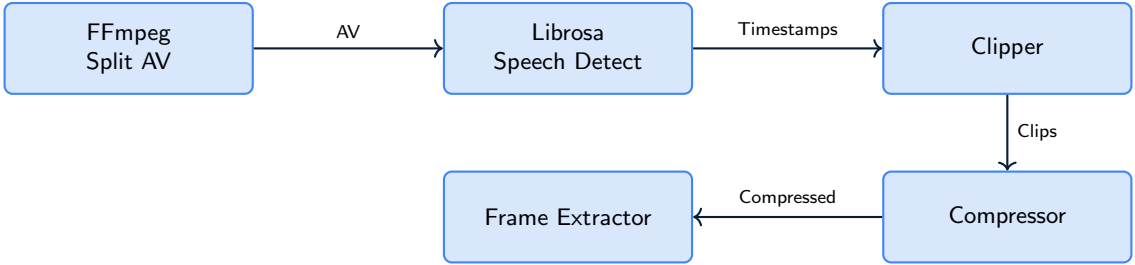

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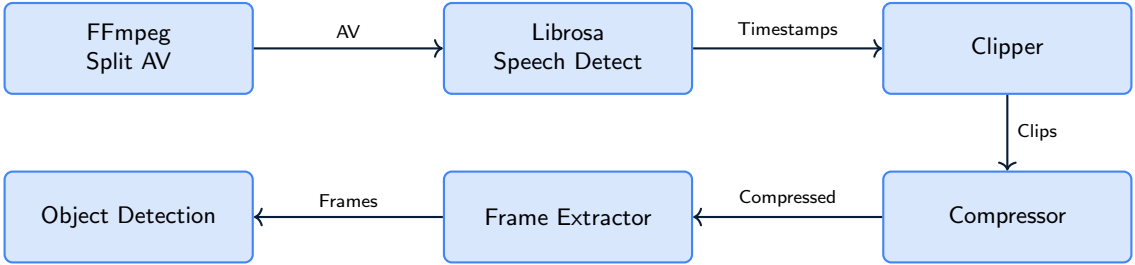

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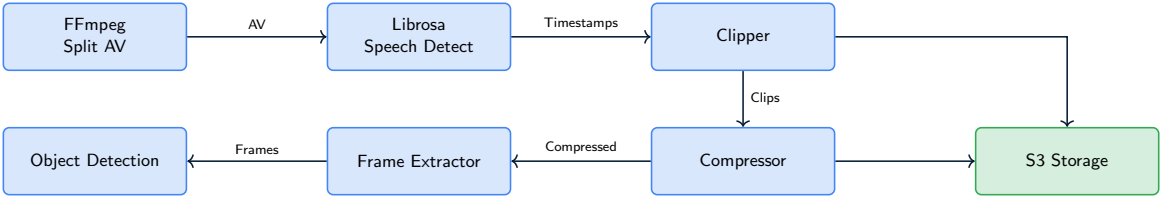

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- Managing this fan-out behavior was important to ensure correct scaling and data flow.

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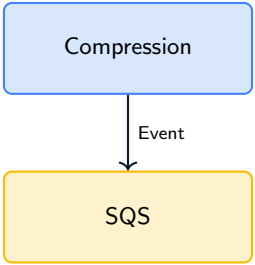

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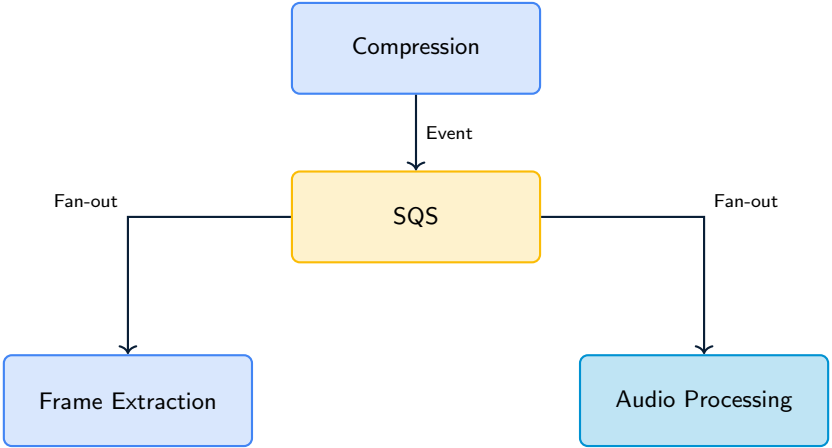

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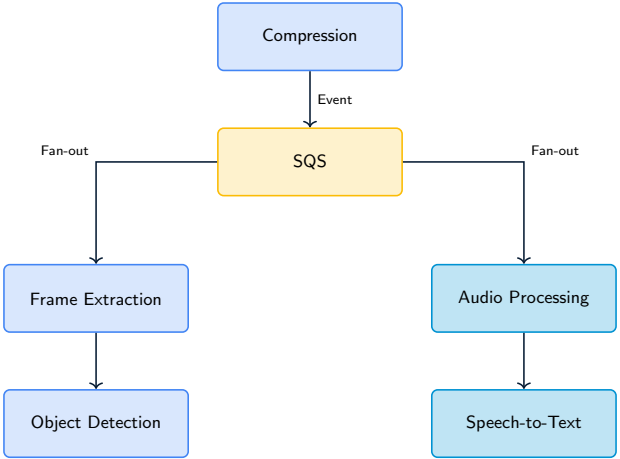

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- I also built wrapper scripts that convert HTTP requests into CLI arguments.

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  - And how I implemented queue-based fan-out execution

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- **AWS ELB Exposure**: Created a NodePort/LoadBalancer service bridging internal Kubernetes networking to an Internet-facing AWS Elastic Load Balancer.
- **Deployment Pipelines**: Utilized `faas-cli deploy -f stack.yml` pointing perfectly to the remote ELB Gateway to construct and map Docker containers directly into EKS Pods.

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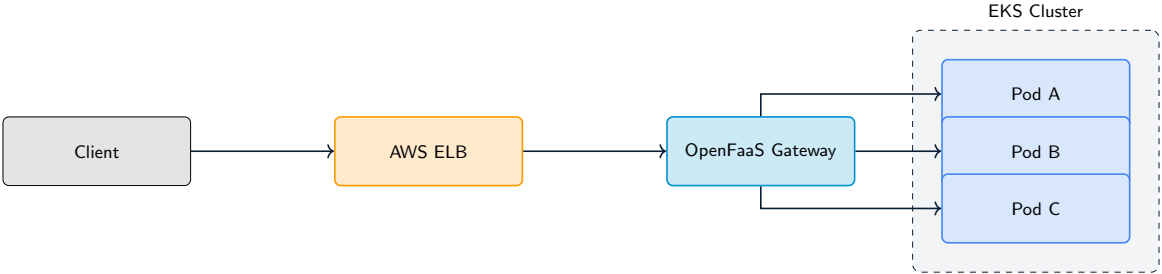

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- I removed dependency on local files by introducing S3-based input/output handling.
- All configuration was centralized into `stack.yml`, making deployment much cleaner.

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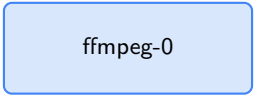
- This made the system fully event-driven and much more scalable.

SQS Orchestration: Flow Topology



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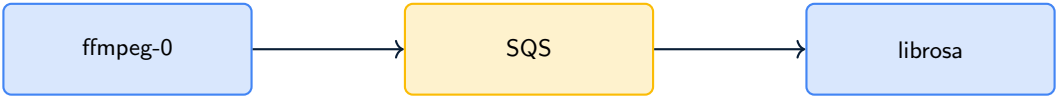

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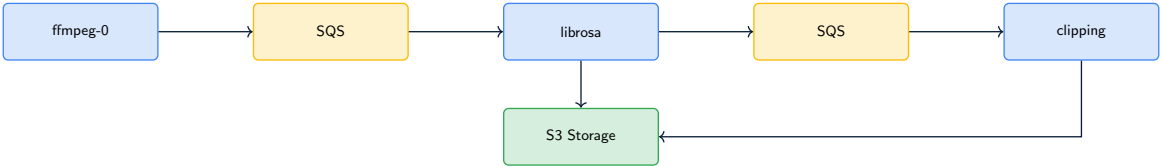

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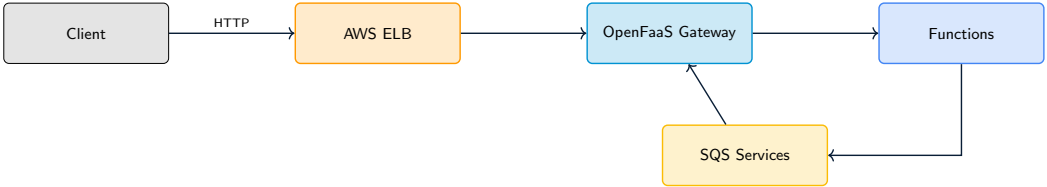

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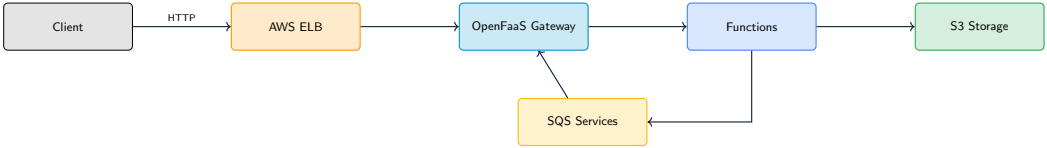

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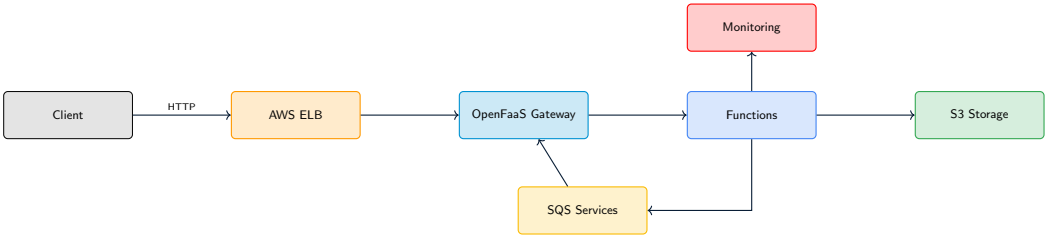

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Logical System Architecture

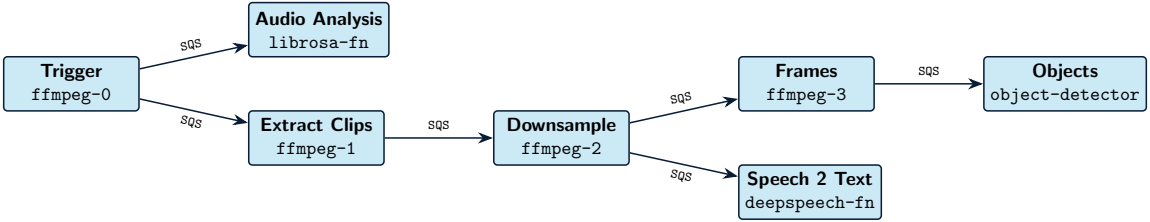

Figure 7: Event-driven OpenFaaS pipeline reliant on S3 mapped payloads.

Key Fixes and Improvements

Major Issues I Solved

- I implemented Locust alongside JMeter to compare performance under different load models.

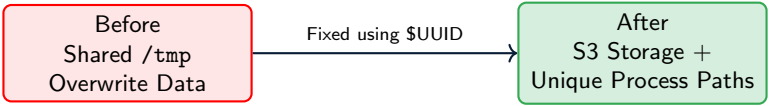

Figure 8: Fixing data overwrites using S3 URIs across parallel invocations.

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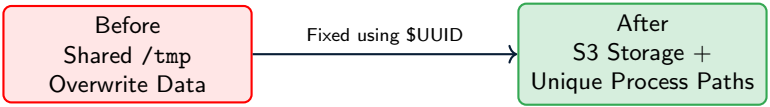

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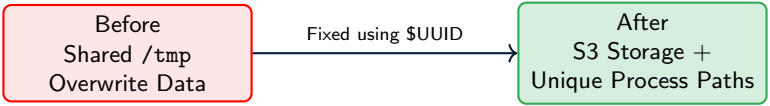

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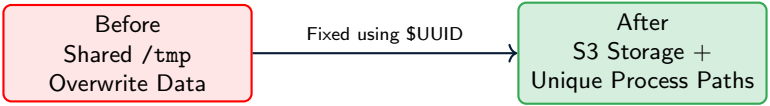

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Visual Architecture: Load Testing

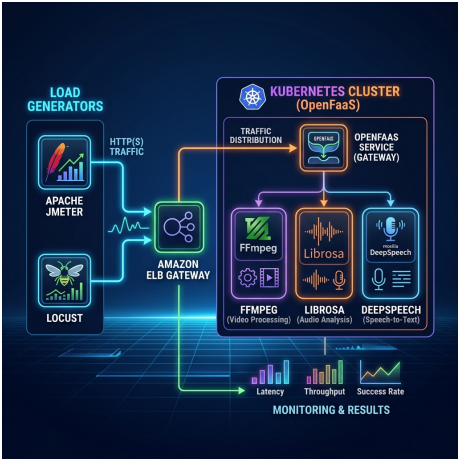

Figure 9: Load Testing Architecture Layout.

Testing Methodology

We simulated user-workloads sending requests to the system utilizing two different load generation mechanisms: **Apache JMeter** and **Locust**.

Test Configuration Pipeline

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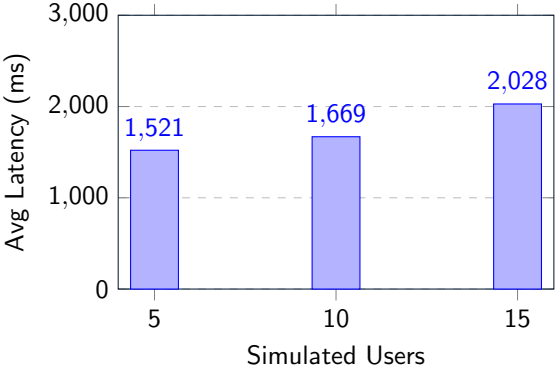
Results: Apache JMeter

Users	Runs	Avg (ms)	Max (ms)	Fail
5	61	1521	1791	0%
10	127	1669	2928	0%
15	202	2028	4226	0%

Table 1: JMeter execution metrics.

- Handled continuous scale perfectly without degradation.



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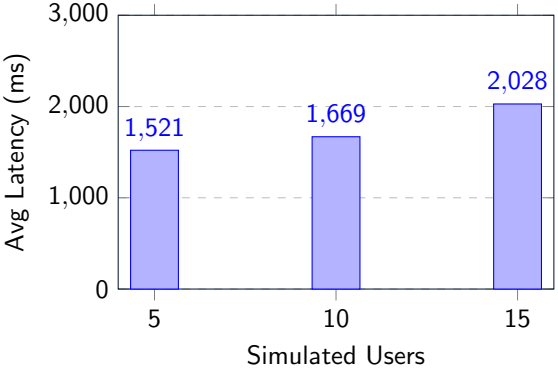

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Results: Locust Framework

Users	Runs	Avg (ms)	Max (ms)	Fail
5	75	1494	1881	0
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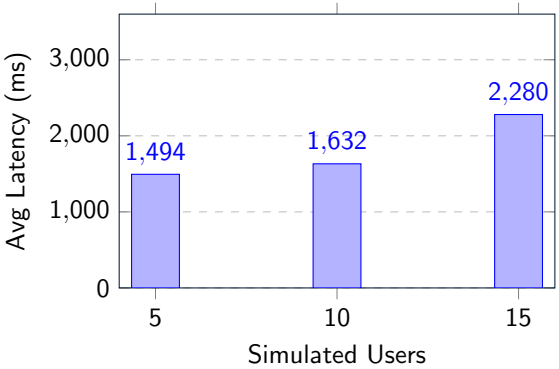
Table 2: Locust metrics.

- Python-based model logged higher tail latency.



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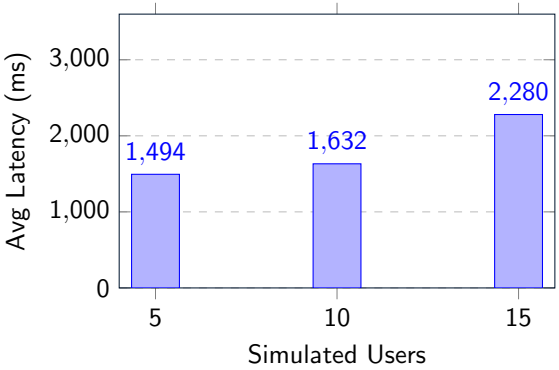

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- **Learning:** This project helped me understand real-world challenges like distributed execution, concurrency issues, and cloud deployment.